

Vanier College
Faculty of Science and Technology
Department of Computer Science

Course title: Internet of Things (IoT)

Teacher: Samad Rostampour

Course Number: 420-531-VA

Section: 3

Teaching Method: In-person

Email: Through MIO (24h delay)

Semester: Fall 2026

Office: K-305

Ponderation: 2-4-3

Office Hours: By appointment

Course prerequisites: All program courses from semesters 1-4

Weekly Schedule: as per advertised on LEA

TEACHER AVAILABILITY

Students can send their questions through MIO and will receive responses with 24h delay (Weekdays).

Note: For sending MIO messages, students must mention the course name and section in the subject of the message. Messages without this subject format will be responded to with a delay.

To make an appointment, students should send their requests at least 3 days in advance.

COURSE DESCRIPTION

From residential systems to industrial controls, the Internet of Things (IoT) is increasingly integrating the physical world into computer-controlled systems. In IoT systems, microcontroller-driven sensors such as thermometers and RFID tags can capture physical data. This data may be sent to processing, storage, and message queuing single-board computer servers, which relay the data to appropriate subscribers or send messages to APIs such as text-messaging services. Subscriber actor devices, such as electrical relays and computer-controlled motors, then translate received messages to actions in the physical world. These systems may be controlled through Web or mobile interfaces and use input from third-party APIs, such as weather in their decision-making processes.

In this course, students learn about the different components, applications, and how to make IoT systems work. They will learn how to select the right IoT technologies, configure, and program them into functional systems. As a final capstone project, students apply the acquired skills to design, implement, and test an original IoT system. They will present their products and demonstrate their usability.

COURSE'S ROLE IN THE PROGRAM

This is a capstone course that helps students integrate different application domains in an IoT development project built of heterogeneous elements of hardware and software that connect the virtual and physical worlds.

This course leverages the material learned in the Web Programming, UNIX, Networking, and Application Programming courses to produce an original project in a team setting and in preparation for the internship. This course applies programming to relevant fields, such as hardware integration.

STATEMENT OF COMPETENCIES

The competency *00SX: Develop applications for connected objects* will be addressed comprehensively in this course.

COURSE-LEVEL LEARNING OUTCOME

Students will design and implement a complete basic IoT system that connects the virtual to the physical world.

KEY LEARNING OUTCOMES

- Write a technical document detailing the different aspects of the project to be delivered, based on the project requirements.
- Identify and analyze key challenges during development, consider multiple solutions, and justify the selected approach.
- Design and implement the system user interface.
- Connect and configure hardware components.
- Program the connected-object application logic and the control or monitoring application logic.
- Perform behavioural testing, calibrate, and debug the connected objects and applications.
- Present and demonstrate the system prototype running in a physical environment.
- Produce the user manual.

TEACHING METHOD

The course will use the following means to develop and evaluate the above topics:

Lecture period

The two-hour lecture is used to introduce course topics.

Weekly Quizzes will be taken in the lecture session. A quiz is a simple exam (1 or 2 questions).

Note: As a notification for minor assessments, quizzes will be held from **Week 3 to Week 14**. Maybe due to the time limitation, in a session, the quiz is postponed but by default, students should be ready.

Related to the lecture content, students will receive two different types of lab activities:

Semester project: As a team, students will build an IoT-based system that will create and manage a smart environment. This project has different phases, and their details (Tasks, due date, etc.) will be sent to the students phase by phase.

Research-based projects: These cover some important topics that cannot be discussed (due to time limitations) during the lecture time and are necessary for a better understanding. All students have to create a PowerPoint file as the output of each research-based project, and during the semester, each student may have one presentation in the class.

Lab period

A four-hour lab allows students to work on lab assignments, solve exercises, and do information research. Lab exercises and assignments will be used to assess the understanding of the material covered.

All lab assignments must be completed and submitted at the lab under the teacher's supervision. If an assignment may be completed outside the lab, the teacher will indicate this. Therefore, submitting a lab assignment without the teacher's confirmation is not acceptable.

COURSE EVALUATION PROCEDURES AND GRADING SCHEME

Description	Grade Value
Weekly Labs Activities and Quizzes	25% (schedule below)
Research-based projects	10%
Written Exam	30%
Semester project (LIA)	35%
Phase 1: Full Functionality* (2%)	
Phase 2: Full Functionality (5%)	
Phase 3: Full Functionality (3%)	
Phase 4: Implementation and functionality, being user-friendly, facilities and logic of the system – Phase 2, 3 and 4 (15%)	
Presentation and final report (10%)	

*: Full Functionality means all parts of this phase of the project should work, and partial demonstration is not acceptable

LIA demonstrations do not have a late submission.

1. Students will be informed of the exact date of each major test (e.g., written exam and semester project, except quizzes) at least one week in advance.
2. To pass the course, students must submit all assignments and obtain a passing average of 60% marks for each of the assignments/demonstrations, the LIA Project and the Final exam. As a rule, **no makeup exam** will be given.
3. Students must participate in each demonstration session. **The first absence will result in removal from the group, and the student will be required to continue the project individually.** The second absence will result in a zero-contribution score for the final project and a zero grade for the LIA project.

4. Since all groups must demonstrate during the same session, any student absent during their group's assigned time slot will be recorded as absent regardless of the session time.
5. Assignments should be submitted on time. Late submissions are not accepted. **The late submission feature on the LEA is accepted only for permitted students.** If the teacher permits it, a 10% deduction applies per day. After a 5-day delay, the submission was not accepted. Please do not send your late assignments via MIO. Students will be expected to answer questions related to the assignment being assessed.
6. For weekly lab activities, students are provided with all necessary materials and instructions. They only need to perform the lab. Therefore, only the final result matters, and there are no specific grades for it.
7. Students are responsible for saving their work using the appropriate media available in the lab.
8. For certain sections of the final project, students are permitted to use code and libraries from the internet or Gen-AI applications.
9. However, for the project design, system integration, and main components of the project, students are not permitted to use code or materials from previous students or Gen-AI tools. The project design and implementation must be unique to each group. Any similarity in these sections will be considered cheating.

Most Important semester events:

- Phase demonstrations (each phase)
- Final demonstration for Phase 4, covering Phases 2, 3, and 4
- Final project presentation — a group activity with a PowerPoint presentation
- Final project documentation — an individual deliverable (final project report)

REQUIRED MATERIALS:

Below are two lists outlining the hardware and software requirements. However, additional products may be introduced during the semester.

Required Software

- Raspberry Pi Operating System
- Node-RED
- Arduino IDE
- ClearStream RFID
- Fritzing

Online Resources

- [Pypi.org](https://pypi.org)
- [Nodered.org](https://nodered.org)
- [Rfidjournal.com](https://rfidjournal.com)
- [Raspberrypi.org](https://raspberrypi.org)
- [Arduino.cc](https://arduino.cc)

Required Hardware

The required materials for the IoT course are listed as follows. The students should prepare them before starting the semester. The links only show the details and students can purchase from any store.

- **Raspberry Pi 5 8GB - 64GB Edition of OS** (https://www.amazon.ca/iRasptek-Starter-Kit-Raspberry-Pre-Installed%EF%BC%88Red/dp/B0D95QBKJ4/?encoding=UTF8&pd_rd_w=OFSzH&content-id=amzn1.sym.c9686649-ca7d-456d-a0b6-a3b312ccfb77%3Aamzn1.symc.a68f4ca3-28dc-4388-a2cf-24672c480d8f&pf_rd_p=c9686649-ca7d-456d-a0b6-a3b312ccfb77&pf_rd_r=32KEN84WV6S84QGK6Q46&pd_rd_wg=Hx4N&pd_rd_r=8884bf90-0f35-40a3-9535-1f1123c35dcb&ref=pd_hp_d_atf_ci_mcx_mr_ca_hp_atf_d&th=1)
- **(Raspberry Pi 3 or 4 with 64-bit OS are also acceptable, but the full kit includes the adaptor, cables, case, and memory card)**
- **Arduino UNO** (https://www.amazon.ca/Elegoo-Board-ATmega328P-ATMEGA16U2-Arduino/dp/B01EW0E0UU/ref=sr_1_1_sspa?crd=EOY8LE2SHRUC&dchild=1&keywords=arduino+uno&qid=1634825150&srefix=arduino%2Caps%2C168&sr=8-1-spons&psc=1&spLa=ZW5jcnlwdGVkUXVhbGlmaWVyPUEwXSIMxN05EU1lWRUpNjMvY3J5cHRlZElkPUEwMTAxODc4MjJCSDJBtRkRaSU9ZVlZlbnNyeXB0ZWRBZEIkPUEwMDE4OUI0TU9TUDFTWFpNN1Imd2lkZ2V0TmFtZT1zcF9hdGYmYWN0aW9uPWNsaWNrUmVkaXJlY3QmZG9Ob3RMb2dDbGljaz10cnVl)
- **ESP-32 Board** (https://www.amazon.ca/Freenove-ESP32-WROOM-Compatible-Wireless-Detailed/dp/B0C9TGJRPH/ref=sr_1_1_sspa?crd=36V1XMSM5XBTO&dib=eyJ2J2ljojMSJ9.a1IfNGmnc5ZAlhtfsdlT0xA5MotbOA-jBO2IJm9md2ltfBjeZgUbLAWKjEW984JmK0WrOw70b50Byl-IW2tWmKDQ8QALffuGJTtlepEIVYfNG1H8MkpkdOBd5bFEUgW0aSQKCN2-KkxHVZ5JKHaiks_qWO4FNjB55gpXvPIS_15cVq8G5lmtR_a4o7V9ZuXIgmZcYhTs1jrlfS_QhNO_Gtz7OZnhVECj6tqhDX180su8dPBCZArruRByny57sQnRx1yA20WpZ8Vg4I6xai-DLMqbB8RwjU6Ie9W8nFqGGCE.58AgeXkAm66pxRDZx8Hlwh7kn7oxq4SvQhUb8T-mKWY&dib_tag=se&keywords=esp32&qid=1723380481&srefix=esp32%2Caps%2C328&sr=8-1-spons&sp_csd=d2lkZ2V0TmFtZT1zcF9hdGY&th=1)
- **Freenove RFID Starter Kit** (https://www.amazon.ca/Freenove-Raspberry-Processing-Tutorials-Components/dp/B06VTH7L28/ref=sr_1_1_sspa?dchild=1&keywords=rfid+kit&qid=1634827993&sr=8-1-spons&psc=1&spLa=ZW5jcnlwdGVkUXVhbGlmaWVyPUEwXSIMxN05EU1lWRUpNjMvY3J5cHRlZElkPUEwNTc4OTY5M0dUNFU1UktRNEtLJmVUy3J5cHRlZEFkSWQ9QTA0MzU1MzQzVDdLVDI1MERVSjE1JndpZGdldE5hbWU9c3BfYXRmJmFjdGlvbj1jbGlja1JlZGlyZWNOJmRvTm90TG9nO2xpY2s9dHJlZQ==)

LEARNING INTEGRATION ASSESSMENT (LIA)

In a group of 2 or 3 students, create and manage a smart environment. The project implementation is divided into milestones. For each milestone, the team of students completes a deliverable consisting of one or more components of their IoT system or from the other project components. At the end of the project, students individually complete their reviews of their peers.

Components of the system:

- Data capture from the environment
- Interconnection of programmable devices such as microcontrollers, SBCs, computers, and mobile devices
- Local or remote data hosting (storing and serving) and processing
- Acting and making decisions based on the currently available data
- Secure the system and preserve privacy, e.g., through authentication

Other project components:

- Behaviour-driven development and testing
- Application presentation and report

Individual work: Assess the involvement of other team members through peer reviews

LIA EVALUATION CRITERIA

- Correctness of the solution
- Usability of the interface
- Reactiveness of the system
- Quality of the coding and software architecture
- Quality of testing and absence of software bugs
- Security of the application
- Quality of the user manual and technical documentation
- Assessment of individual performance through peer evaluations
- Evaluation of individual contribution through, e.g., regularity of and contribution to version control system commits

SPLI

The quality of English expression will be evaluated and is worth 10% of the total as part of grading the LIA written reports and in-class presentations.

Course withdrawal and drop deadlines

Students have until September 19, 2025, to withdraw from a course without penalty or added remark on their transcript. Students who drop the course between September 20 and November 3, 2025, will receive no numerical grade but an AE notation on their transcript, which indicates that they have dropped the course after the September 19 deadline.

COLLEGE POLICIES & PROCEDURES

It is your responsibility to be aware of the various policies and procedures governing your rights and obligations while you are attending Vanier College.

It is the student's responsibility to be familiar with and adhere to all Vanier College Policies and potential modifications due to the COVID-19 situation.

A summary of the course-level policies that apply in this and in all other Vanier courses can be found under "Course-Level Policies" in Important Vanier Links on Omnivox, or by following this link: <http://www.vaniercollege.qc.ca/psi/course-level-policies/>.

Complete policies can be found on the Vanier College website, under [Policies](#).

Tentative Weekly Plan (could be changed based on unpredictable situations)			
Week	Theory (2 hours)	Lab (4 hours)	Homework (3 hours)
1	<i>Internet of Things (History, concept and applications)</i>	<i>Installation of software, preparation, and exploration of the environment</i>	<i>Completion of the lab</i>
2	<i>IoT system layers, structures (e.g., Capture, data transmission and application)</i>	<i>Looking at an existing system. Hello World program on Raspberry Pi (and on Arduino).</i>	<i>Research-based assignment based on the current IoT system.</i>
3	<i>Single-board computers and micro-controllers (Raspberry Pi, Arduino, Nvidia Jetson, etc.) - Part 1</i>	<i>Using the GPIO to connect the sensors and making the software capture to match for two devices separately.</i>	<i>Four-week assignment</i>
4	<i>IoT actors and sensors (With Raspberry Pi) – Humidity-Temperature, Photo-resistor, DC and Stepper Motors</i>	<i>Arduino IDE, Arduino Programming</i>	<i>continued</i>

5	Single-board computers and micro-controllers (Raspberry-Pi, Arduino, Nvidia Jetson, etc.) - Part 2	Implement a scenario for input on one device leading to output on another device with publish-subscribe (skipping network details). (Start Phase-1)	Start the semester project (LIA) Scenario phase 1: Creating a smart place. Week by week, complete a part of the project
6	IoT actors and sensors (With Arduino)- Humidity-Temperature, Photo-resistor, DC and Stepper Motors	continued	continued
7	IoT platforms and communication protocols (e.g., IBM Node-Red, Azure, MQTT, etc.)	Node-Red. Connect the devices and run MQTT. (Start Phase-2)	Demonstration of Phase 1 and start Phase 2
8	Wi-Fi technology (Antennas, Bands and Frequencies, applications)	Configure a Wi-Fi adapter. Capturing and manipulating data (maybe edge computing).	continued
9	Bluetooth (Version, hardware components)	Now with Bluetooth.	Deliverable: Plan for data collection and processing due.
10	RFID (Standards, RFID Active, Passive, components)- Part 1	NFC RFID – Very short-range proximity identification, e.g., access control and identification for the smart Home - (Start Phase-3)	Demonstration of Phase 2 and start Phase 3.
11	RFID (Standards, RFID Active, Passive, components)- Part 2	UHF RFID – warehousing and stores (smart shelves, smart gates)-	continued
12	Image processing-based technologies (e.g., Barcode and Camera) - Part 1	Barcode identification, esp. QR code, invisible barcode (UV) (Start Phase-4)	Demonstration of Phase 3 and start Phase 4
13	Image processing-based technologies (e.g., Barcode and Camera) - Part 2	Simple code for object detection, e.g., counting people in an area.	Testing of the project – creating a testbed – due date.
14	IoT Security: Privacy vs. Security	Bad scenarios with unauthorized access and the solution (securing the systems)	Demonstration of Phase 4, which includes the entire project
15	Written Exam	Semester Project presentation and evaluation	

Course withdrawal and drop deadlines

Students should consult the academic calendar for the specific Course Withdrawal and Course Drop deadlines. These are two deadlines to stop attending courses.

Students who drop the course by the first deadline are cancelling their registration in the course without penalty or added remarks to their transcript.

Students who drop the course after the first deadline but before the second deadline will receive no numerical grade but an AE remark (abandon d'études) indicating that they have dropped the course later in the semester after the first deadline. Part-time student status resulting from a course drop after the first deadline will see no fee adjustments as they will be deemed full-time.

Cheating & Plagiarism

Cheating and plagiarism are not tolerated. It is important to refer to the IPESA Academic Integrity section (11.4), and to note the following:

- The penalty for plagiarism or cheating is a grade of zero for that coursework or assignment, without the possibility of revising that coursework or assignment or preparing an alternate assignment.
- Anyone who knowingly permits his or her work to be copied is considered to be as guilty as the cheater. Thus, students must not share their labs, assignments, or exam work with other students before confirming with the teacher.

Absence for a medical reason

In the case of a missed assessment for a short-term medical reason, the student needs to document the absence with a verifiable medical note (or

other medical documentation deemed legitimate by the College) in order to receive an alternative assessment plan. A student's self-declaration of a

medical absence (<https://bit.ly/medical-absence>) will be considered valid documentation, providing the student follows the appropriate steps of

reporting to their teacher(s) for each day they are absent. If the medical absence is for 5 or more consecutive school days, the student is required to

provide valid medical documentation from a physician to support their situation.

Proposed new blurb for course outlines:

It is the student's responsibility to be familiar with and adhere to the Vanier College Academic Policies. The complete policies can be found on the Vanier College website, under [Policies](#). Students should pay particular attention to the [Institutional Policy on the Evaluation of Student Achievement \(IPESA\)](#). The

IPESA aims to clarify the structures and means that ensure the fair and equitable evaluation of student achievement, including the conditions for learning, assessment, and certification.

Section 11.4 of IPESA provides details on **Academic Integrity**, which is an essential component of any educational institution and all students are expected to uphold the highest standards of academic integrity during their college studies. Failure to do so may result in serious consequences – please see section 11.4.2 on **Academic Misconduct**. There are many [resources](#) available to support academic integrity in your courses. If you have a problem that you have been unable to resolve by talking with your teacher, please reach out to the [Student Advocate](#) or to a Faculty Dean's office (A-286, B-208, N-301). For Continuing Education courses, students can visit E-115.